

HENRY JIANG

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Education

Georgia Institute of Technology

Atlanta, GA

May 2026

- B.S. Computer Science | Concentrations: Artificial Intelligence, Computational Media
- GPA: 3.7/4.0 | Graduating with Highest Honors, Dean's List 2023-2025, Eta Kappa Nu Scholar
- AI@GT, Data Science @ GT, GT Animation Club, Chinese Students Association

Skills

Languages: Python, Java, JavaScript, C, C#, C++, HTML/CSS, Typescript, SQL, R

Tools: Git, JUnit, Jira, Docker, Tensorflow, Pytorch, Matplot, Pandas, Numpy, Selenium, OpenCV

Frameworks: React, Next.js, Node.js, Angular, Figma, PostgreSQL, MySQL, MongoDB, AWS, Firebase, Linux, RESTful API

Experience

GT Fung Research Group

Aug 2023 – May 2026

Undergraduate Research Assistant

Atlanta, GA

- Published thesis for graph neural network-based crystal structure prediction achieving 90%+ accuracy with PyTorch Geometric, ASE, and Pyswarms, and contributed to 3 GNN lab packages
- Engineered data preprocessing pipeline processing 10,000+ crystal structures from Materials Project database, discovering 100+ novel ground-state configurations through automated DFT validation
- Solved inverse design bottleneck for unknown lattice parameters with a hybrid particle swarm optimization algorithm, achieving 80% structure match rate and reducing prediction error by 50% compared to baseline methods

Blender Foundation

Apr 2024 – Dec 2025

Extension Developer & Contributor

- Designed and shipped 2 Blender addons using Python and Blender API, enhancing modeling and animation workflow efficiency by 20% for 6,000+ active users with automated IK/FK snapping and algorithmic topology based selection tools
- Resolved 3 high-priority animation module bugs in collaboration with senior developers via Github and Gitea to optimize weight painting scripts issues, reducing total pipeline open issues by 25%

NASA

Jun 2024 – Aug 2024

Information Systems Security Intern

Kennedy Space Center, FL

- Engineered 4 production Splunk dashboards for Launch Control Center monitoring 100,000+ security events, implementing real-time alert systems with custom SPL queries and automated data aggregation pipelines
- Built automated data preprocessing pipeline for security event classification, achieving 75% reduction in false positive alerts through threshold optimization and time-series analysis techniques
- Drafted 20+ PO&AMs for 3 NE-XC sector teams with cross-collaboration, ensuring NIST/FISMA compliance for SSP and SCP documentation.

Projects

Adversarial Machine Learning Based Social Media | Postgres, React, MongoDB, Next.js, Node.js, Angular, AWS, Figma

- Developed an ML-based image poisoning-focused social media site for artists with React, Next.js, and Node.js ensuring 100% safety from web crawling and 99% reduction in unflagged AI-generated posts
- Engineered a horizontally scalable backend on AWS with PostgreSQL and MongoDB, managing complex state transitions and concurrent session management across up to 100,000+ users

DARWIN: Agentic Evolutionary GPT Training Framework | [Github](#) | [arXiv](#) | Python, LLMs, Docker, Claude API

- Designed Docker containerized agentic evolutionary AI with self-improving agents optimizing NN training code via genetic algorithm architecture achieving 1.2% MFU and 2% perplexity optimization on fully trained models
- Engineered CI/CD infrastructure for dynamic agent request pipeline and persistent JSON memory system, improving CoT response quality by 30% and autonomously increasing dataset size by 400% and architecture depth by 200%

Mandelbrotian Volatility Options Trading Model | [Github](#) | [Quantconnect](#) | QuantConnect, Python, Scipy, Scikit-learn

- Engineered a fractal regime-detection OTM options strategy with Hurst exponents (DFA), Hill tail-index, and multifractal spectral width to classify volatility regimes achieving 55+% CAGR and 1.18 Sharpe across 105 trades over 5 years ('20-'25)
- Built gradient boosting trade filter on 27 technical features with scikit-learn cross-validation, regime-adaptive stop-losses and dynamic position sizing delivering +36% average trade PnL and 1.45+ Sortino across MAG7 tech equities